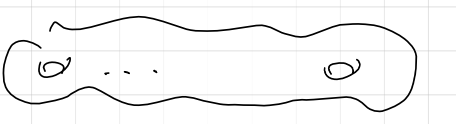
S^n, ~~$S^1\times S^1$~~, $\mathbb{R}P^n$,



$(n\geq 1$
↓
$\pi_1=0.$

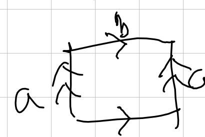
2-dim mfd

$L(p,q)$

$D^1\times S^1 / e^{2\pi i p}, e^{2\pi i q}$

$M \vee N.$ ∧ ∨

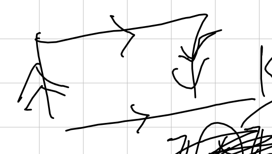
$[0,1]^2$



torus.

$\mathbb{Z}\times\mathbb{Z}$

Klein.



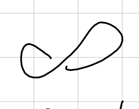
Klein. free product.

~~$\mathbb{Z} * \mathbb{Z}$~~

abababb..

$\langle a,b \mid aba=b\rangle$.

$aba^{-1}b^{-1}=e.$

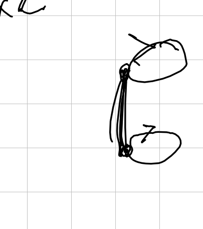


$\langle a,b \mid ab=ba\rangle$

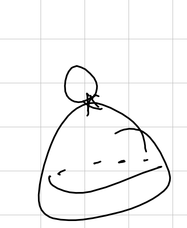
$\mathbb{Z}\times\mathbb{Z}$

$S^1\vee S^1$

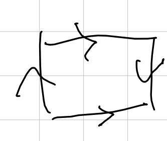
ab=ba → grup 구

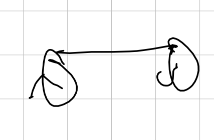


$S^1\vee S^2$



$T: \mathbb{Z}\times\mathbb{Z}$





$S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}$

$T\setminus\bullet: \mathbb{Z}*\mathbb{Z}$

$K\setminus\bullet: \mathbb{Z}*\mathbb{Z}$

theorem

Theorem

$S^1\vee S^1$

$\mathbb{Z}\times\mathbb{Z}$

S^1
↕
$\mathbb{Z}$

$S^3: \underline{0}.$
↓
$\mathbb{R}^3: \underline{0}.$

M: n-dim
with $(n\geq 3)$

then $\pi_1(M)\simeq \pi_1(M\setminus\bullet).$

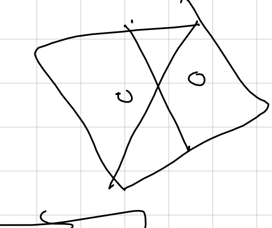
$\simeq \pi_1(M\setminus 2pt).$ $\pi_1(S^n)=0$
$(n\geq 2.$

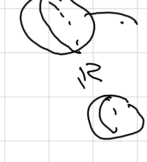
$\pi_1(\mathbb{R}P^n)=\mathbb{Z}_2$
$(n\geq 2)$.

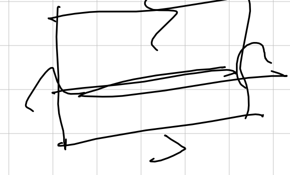
$\simeq \pi_1(M\setminus\{x_n\})$.

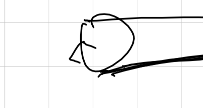
tree

$S^1\vee S^1\vee\mathbb{R}.$
↓
$\mathbb{Z}*\mathbb{Z}$



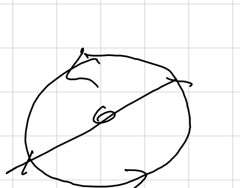


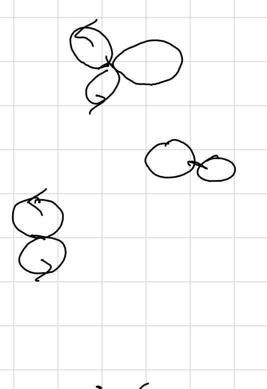




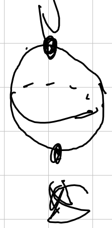
$: S^1\vee S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$

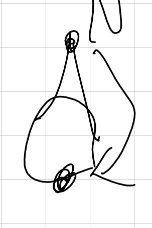


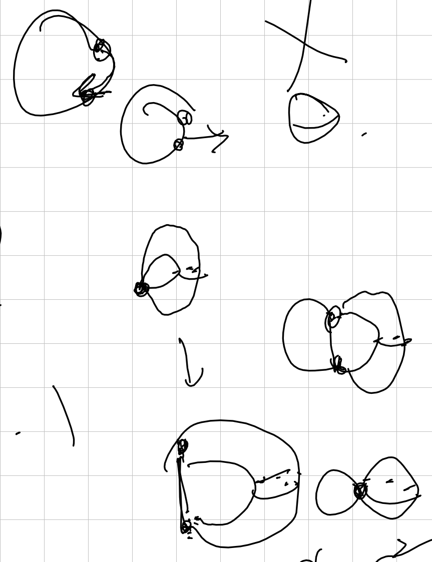


Exemple.



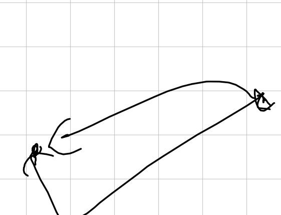
$S^2/N=S$

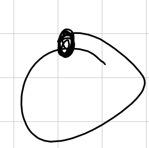




→ $\pi_1(\ .\ .\ .\ )$

$S^1\vee S^2$





S^1: 0-cell 1개
1-cell 1개.

S^2: 0-cell 1개
2-cell 1개

$S^1\vee S^1$

$\mathbb{R}P^2$: 0-cell: 1개
1-cell: 1개
2-cell 1개

$aba^{-1}b^{-1}=0.$

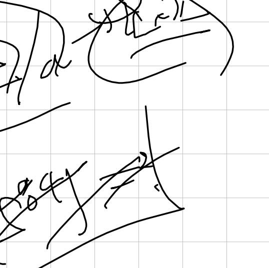
$\langle a,b \mid ab=ba\rangle$

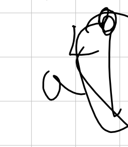
$(\mathbb{Z}*\mathbb{Z})/\alpha$

$\mathbb{R}P^2: \mathbb{Z}_2$ $\langle a \mid a^2\rangle$
$a^2=1$

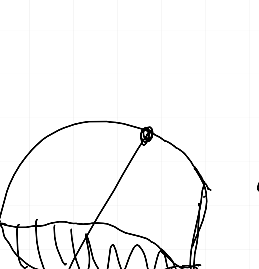
$\mathbb{R}P^3$: ~~0-cell 1개~~
~~1-cell 1개~~
~~2-cell 1개~~
~~3-cell 1개~~

~~$\pi_1($[illegible]$)$~~



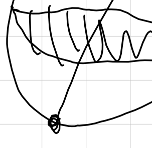


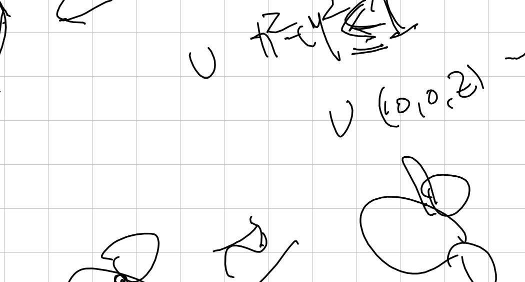
$\pi_1(\mathbb{R}P^4)=\pi_1(\mathbb{R}P^2)=\mathbb{Z}_2.$

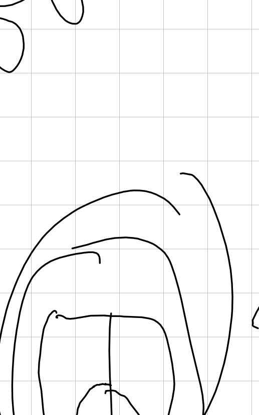


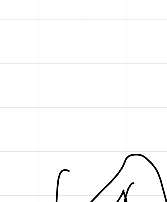
← $x^2+y^2+z^2=1$
∪ $x^2+y^2\leq 1$
∪ $(0,0,z)$ $-1\leq z\leq 1.$

$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$





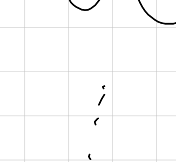


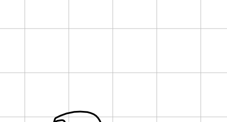


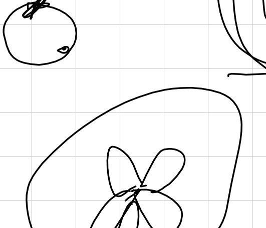
?

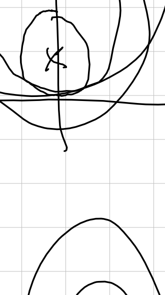
2

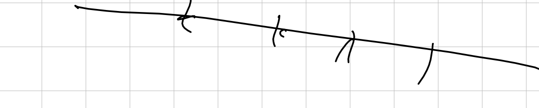
$\mathbb{R}/\mathbb{Z}$



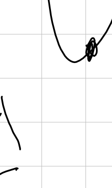








$P=0$



$z=x^4+y^4$
↓ exp

$K_1=f(4,2)$
$K_2=f(4, \ldots$

$\frac{1}{r^2+z}.$

$\int \frac{r\,dr}{r^2+z}$ $\int \frac{1}{z}\,dz$
$\frac{1}{4}$
$2\pi i$

$2\pi \cdot X(\cdot)^{-1}$
1
0